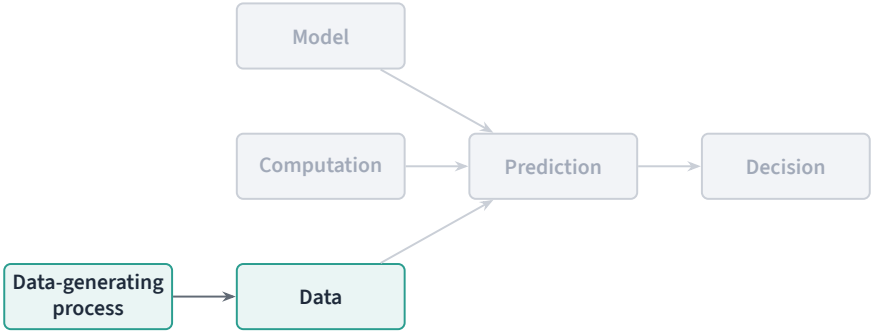
ACCELERATE PROGRAMME FOR SCIENTIFIC DISCOVERY

Uncertainty in AI

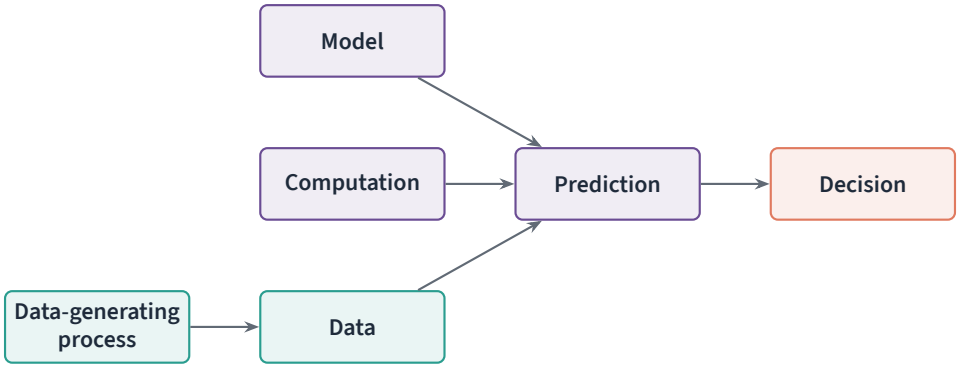
So far



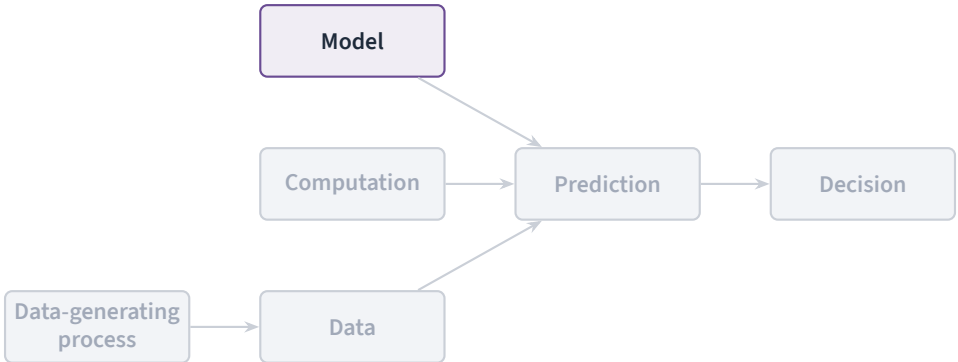
data-generating process may be random: $p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$

data samples from it: what, how and how much we measure

Next: the model

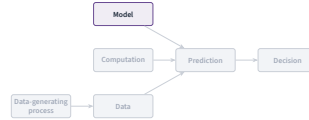


Next: the model



What to model, what kind of model, and which settings of it

A model of reality

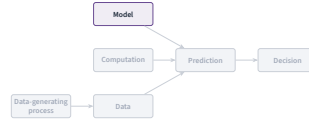


“As far as the laws of mathematics refer to reality, they are not certain; and as far as they are certain, they do not refer to reality.”

— Albert Einstein

A. Einstein (1921). Geometry and experience. Lecture, Prussian Academy of Sciences, Berlin.

A model of reality



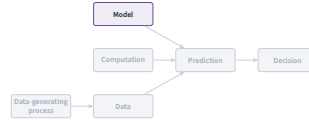
“As far as the laws of mathematics refer to reality, they are not certain; and as far as they are certain, they do not refer to reality.”

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A model of reality cannot be certain, so we make it probabilistic

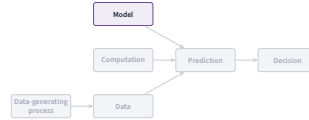
A. Einstein (1921). Geometry and experience. Lecture, Prussian Academy of Sciences, Berlin.

A model is a distribution



von Neumann: a model is a mathematical construct describing what we observe

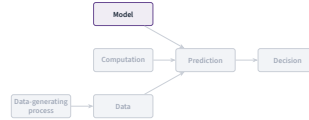
A model is a distribution



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A model is a distribution

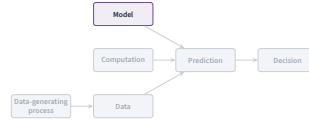


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so: **our model is a probability distribution**

A model is a distribution



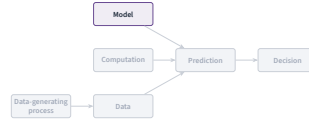
von Neumann: a model is a mathematical construct describing what we observe

Einstein: about reality, it cannot be certain

so: **our model is a probability distribution**

just as the true process was a joint distribution $p_{\text{true}}(x, y)$, our model is a joint distribution over everything we will measure

A model is never exactly right

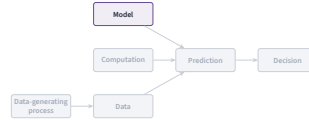


“All models are wrong, but some are useful.”

— George E. P. Box

G. E. P. Box and N. R. Draper (1987). *Empirical Model-Building and Response Surfaces*, p. 424.

A model is never exactly right



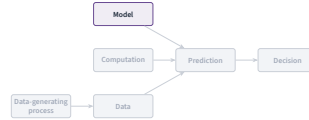
“All models are wrong, but some are useful.”

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Our model will be wrong too; the question is whether it is useful

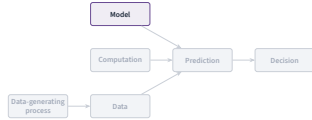
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Our model: a joint distribution



Encodes how the variables relate, and what we assume

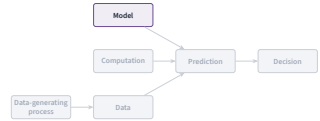
Our model: a joint distribution



Encodes how the variables relate, and what we assume

the data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$

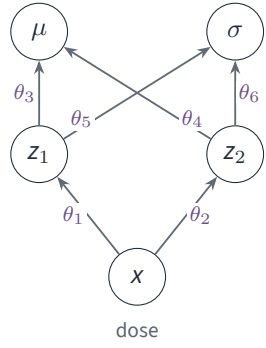
Our model: a joint distribution



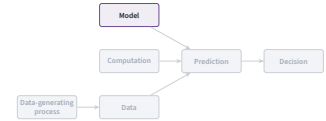
Encodes how the variables relate, and what we assume

the data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$

the settings θ: all the numbers on the network's connections



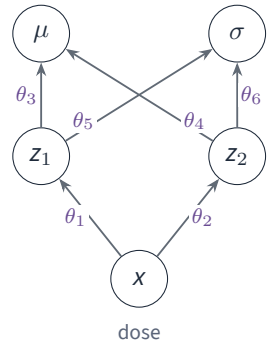
Our model: a joint distribution



Encodes how the variables relate, and what we assume

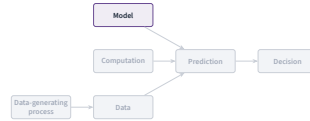
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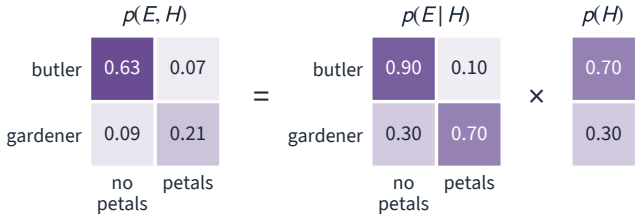
the settings θ: all the numbers on the network's connections



Our model: the joint $p(\mathcal{D}, \theta)$

Building the joint: the product rule

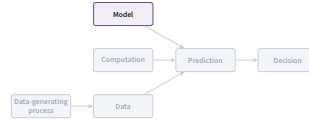




$$p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta)\, p(\theta)$$

as for Holmes: a conditional and a prior

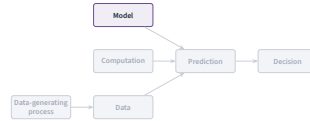
The conditional: assumptions



$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^{N} p(x_i, y_i \mid \theta)$$

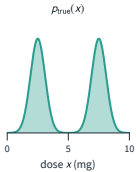
patients are independent, given θ

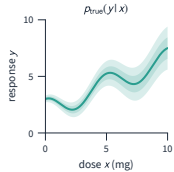
The conditional: assumptions

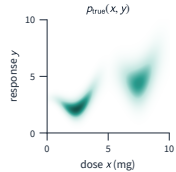


$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^{N} p(x_i, y_i \mid \theta)$$ patients are independent, given θ

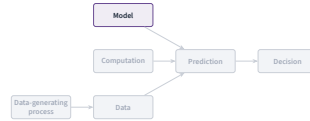
$$= \prod_{i=1}^{N} p(y_i \mid x_i, \theta)\, p(x_i)$$ the product rule, for each patient







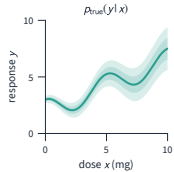
The conditional: assumptions



$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^{N} p(x_i, y_i \mid \theta)$$ patients are independent, given θ

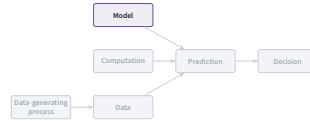
$$= \prod_{i=1}^{N} p(y_i \mid x_i, \theta)\, p(x_i)$$ the product rule, for each patient

$$\propto \prod_{i=1}^{N} p(y_i \mid x_i, \theta)$$ doses are chosen: $p(x_i)$ is free of θ

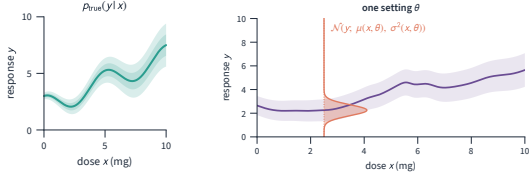


We model only $p_{\text{true}}(y \mid x)$: our conditional is $p(y \mid x, \theta)$

The conditional: assumptions

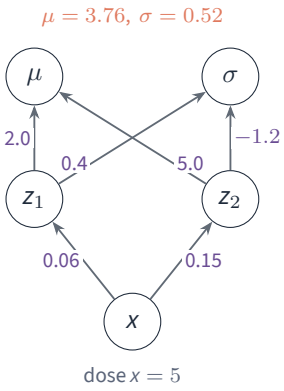


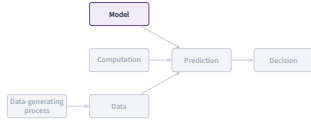
$$
\begin{aligned}
p(\mathcal{D} \mid \theta) &= \prod_{i=1}^{N} p(x_i, y_i \mid \theta) && \text{patients are independent, given } \theta \\
&= \prod_{i=1}^{N} p(y_i \mid x_i, \theta)\, p(x_i) && \text{the product rule, for each patient} \\
&\propto \prod_{i=1}^{N} p(y_i \mid x_i, \theta) && \text{doses are chosen: } p(x_i) \text{ is free of } \theta \\
&= \prod_{i=1}^{N} \mathcal{N}\big(y_i;\ \mu(x_i, \theta),\ \sigma^2(x_i, \theta)\big) && \text{each response: a Gaussian}
\end{aligned}
$$



We model only $p_{\text{true}}(y \mid x)$: our conditional is $p(y \mid x, \theta)$

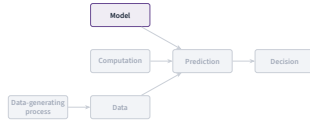
The conditional: a network

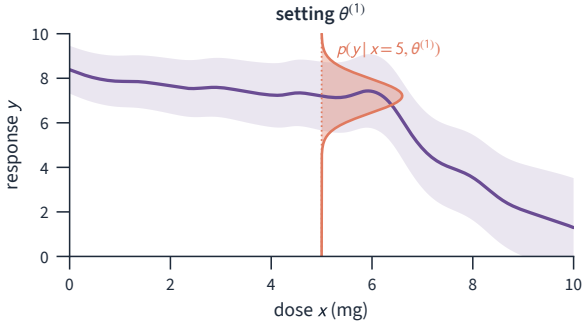




$$p(y \mid x, \theta) = \mathcal{N}\big(y;\ \mu(x,\theta),\ \sigma^2(x,\theta)\big)$$

The conditional: different settings, different bands

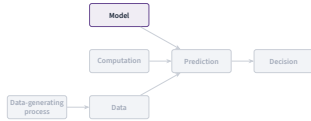


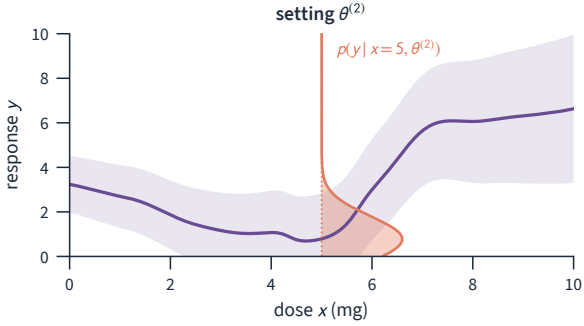


one architecture, different settings θ: each setting gives its own band over all doses

at each dose, a Gaussian: one row of our table

The conditional: different settings, different bands

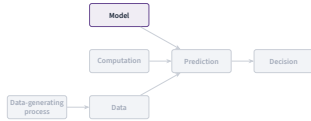


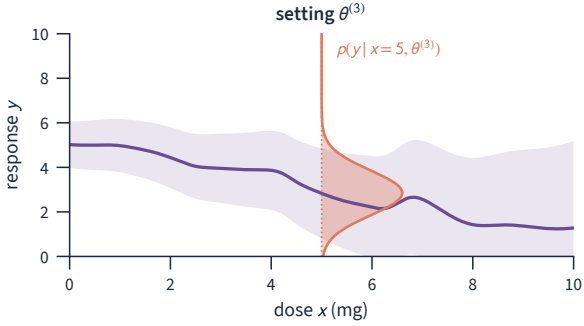


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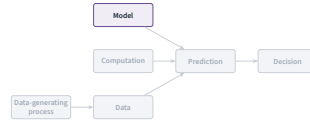




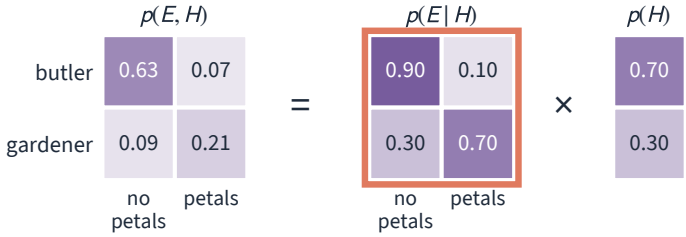
one architecture, different settings θ: each setting gives its own band over all doses

at each dose, a Gaussian: one row of our table

Where are we?

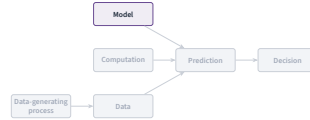


$$p(\mathcal{D}, \theta) = \boxed{p(\mathcal{D} \mid \theta)}\, p(\theta)$$

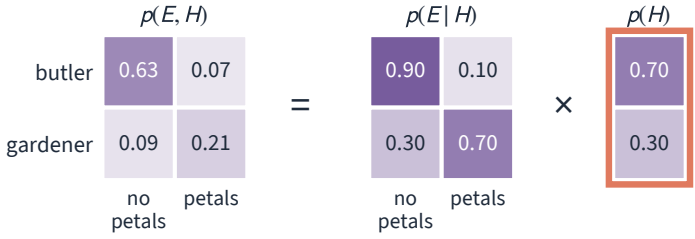


done: the conditional, as Holmes's $p(E \mid H)$

Where are we?

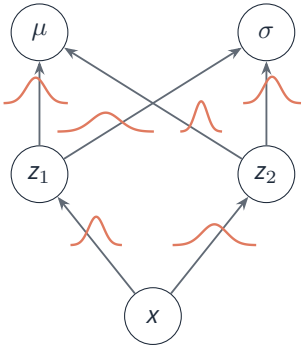


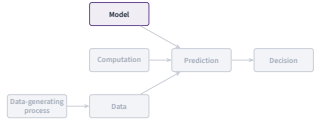
$$p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta)\, p(\theta)$$



next: the prior, as Holmes's $p(H)$

The prior

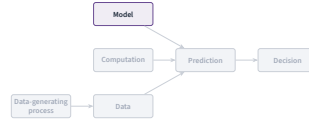


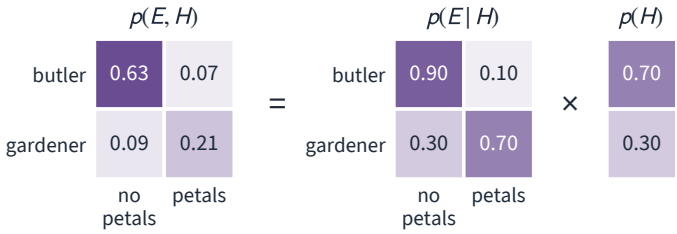


$$p(\theta) = \mathcal{N}(\theta;\, m,\, S)$$

S diagonal: one independent bell per number

Putting it together: the joint

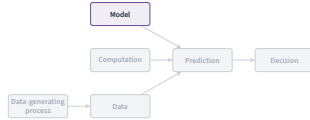




$$p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta)\, p(\theta)$$

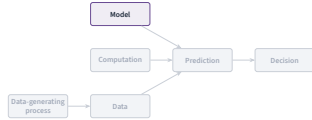
As for Holmes: multiply the prior and the conditional

Predictions



We have specified the joint: $p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta)\, p(\theta)$

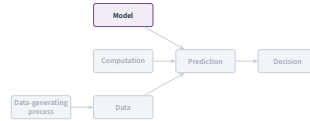
Predictions



We have specified the joint: $p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta)\, p(\theta)$

To predict, add a new pair $(x_\star, y_\star)$: we know the dose $x_\star$, but never observe $y_\star$

Predictions



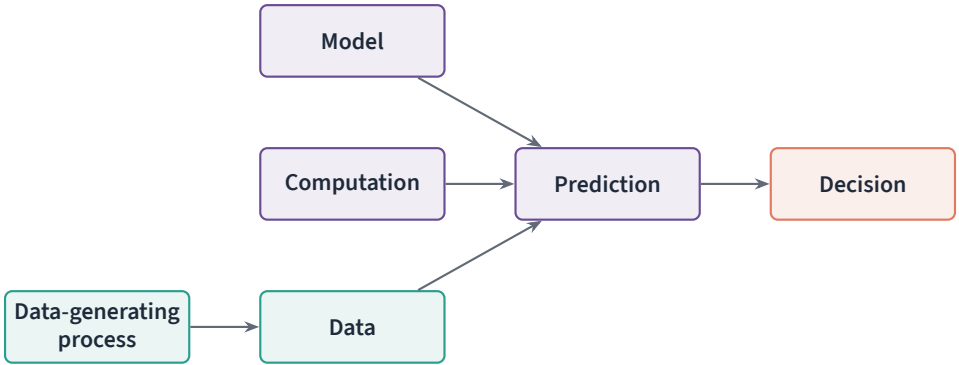
We have specified the joint: $p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta)\, p(\theta)$

To predict, add a new pair $(x_\star, y_\star)$: we know the dose $x_\star$, but never observe $y_\star$

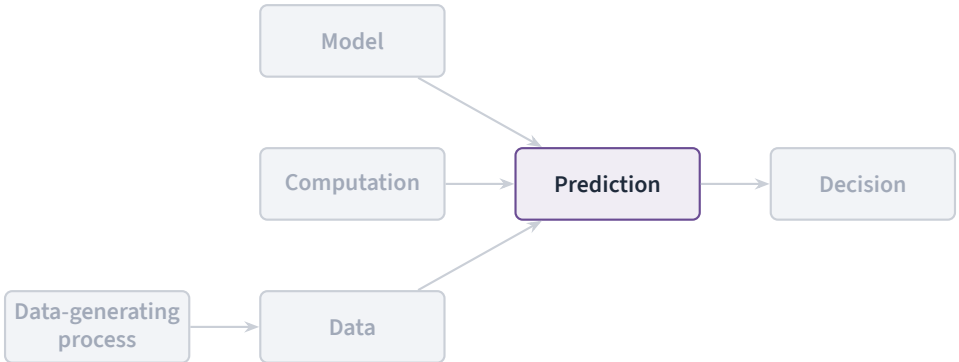
$$p(y_\star, \mathcal{D}, \theta \mid x_\star) = p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)$$

The model already contains the new patient: prediction is just probability, as we shall see next

Next: the prediction

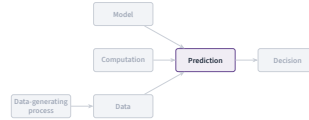


Next: the prediction



Combining data, model and computation

The prior predictive



$$p(y_\star, \mathcal{D}, \theta \mid x_\star)$$

$\downarrow$ add up over θ

$$p(y_\star, \mathcal{D} \mid x_\star)$$

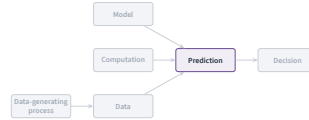
$\downarrow$ add up over $\mathcal{D}$

$$p(y_\star \mid x_\star)$$

Let's predict first without data:

$$p(y_\star \mid x_\star) = \int p(y_\star \mid x_\star, \theta)\, p(\theta)\, d\theta$$

The prior predictive



$$p(y_\star, \mathcal{D}, \theta \mid x_\star)$$

add up over θ

$$p(y_\star, \mathcal{D} \mid x_\star)$$

add up over $\mathcal{D}$

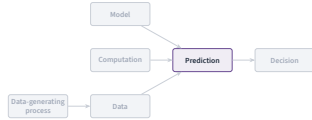
$$p(y_\star \mid x_\star)$$

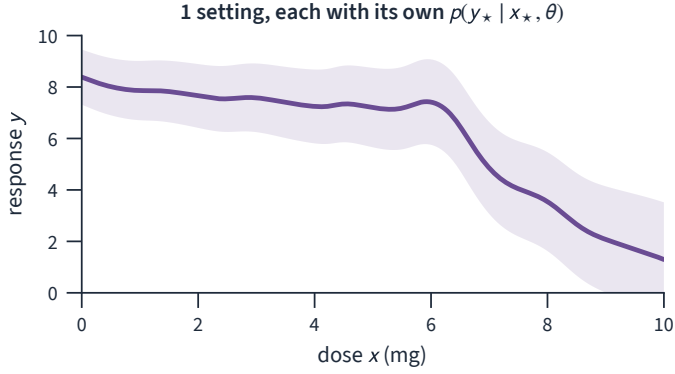
Let's predict first without data:

$$p(y_\star \mid x_\star) = \int p(y_\star \mid x_\star, \theta)\, p(\theta)\, d\theta$$

think of the integral as a sum over all settings: each setting gives its band $p(y_\star \mid x_\star, \theta)$, with weight $p(\theta)$

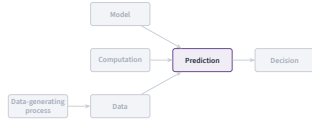
The prior predictive, setting by setting

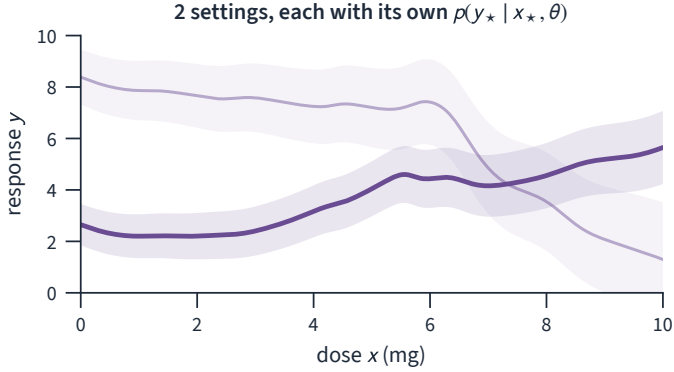




one setting at a time: each gives its own band

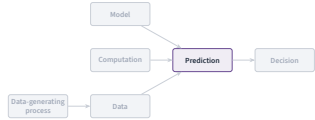
The prior predictive, setting by setting

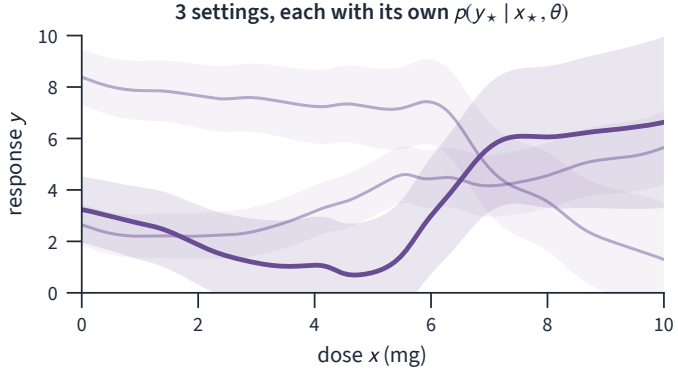




one setting at a time: each gives its own band

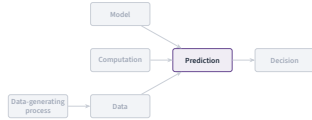
The prior predictive, setting by setting

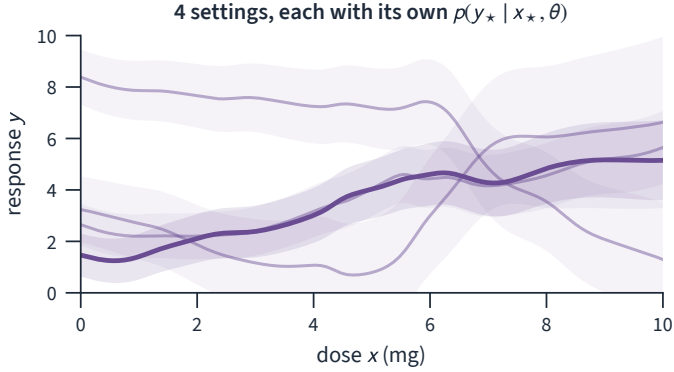




one setting at a time: each gives its own band

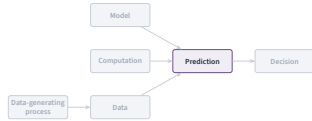
The prior predictive, setting by setting

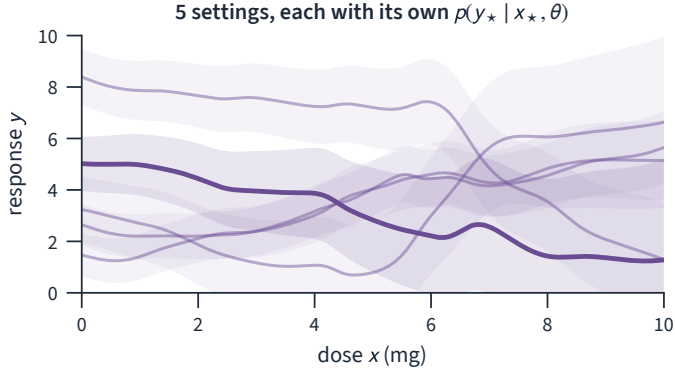




one setting at a time: each gives its own band

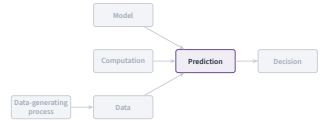
The prior predictive, setting by setting

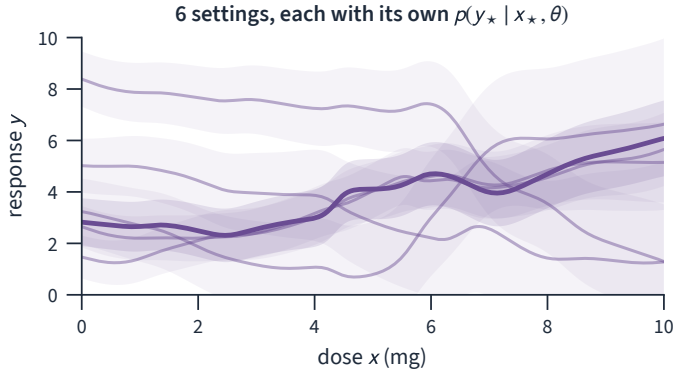




one setting at a time: each gives its own band

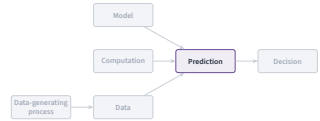
The prior predictive, setting by setting

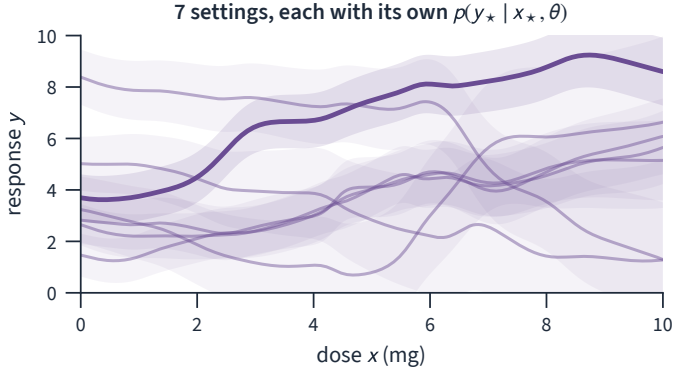




one setting at a time: each gives its own band

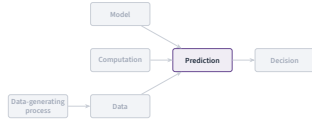
The prior predictive, setting by setting

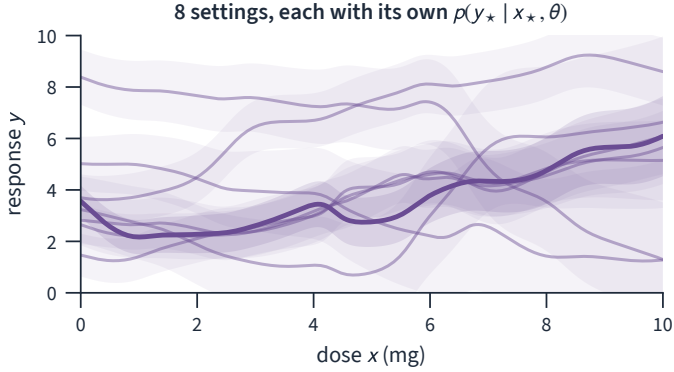




one setting at a time: each gives its own band

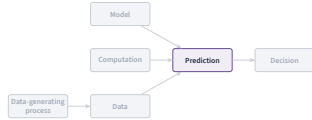
The prior predictive, setting by setting

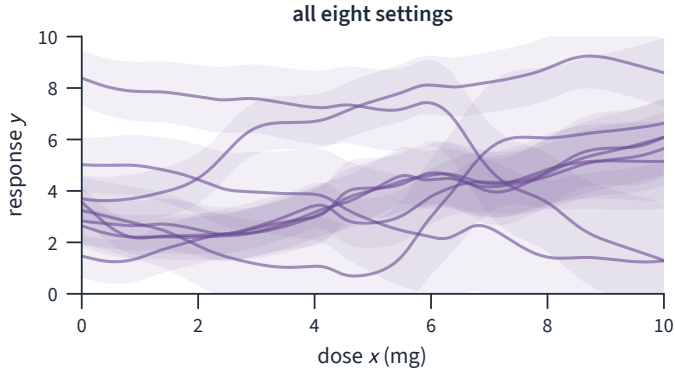




one setting at a time: each gives its own band

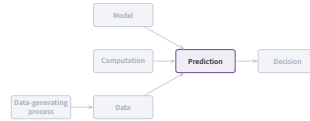
The prior predictive, setting by setting

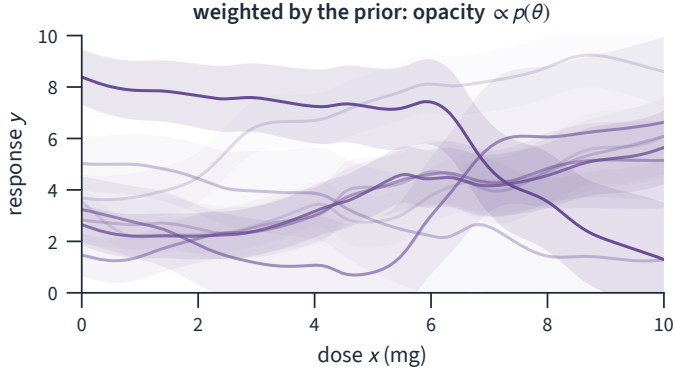




all eight settings

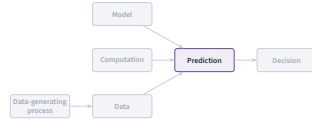
The prior predictive, setting by setting

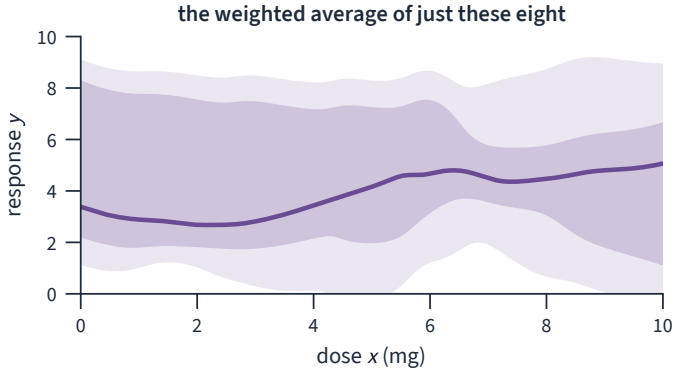




each band's weight in the integral: its prior density

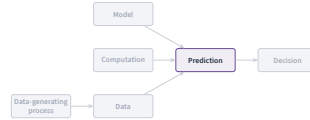
The prior predictive, setting by setting

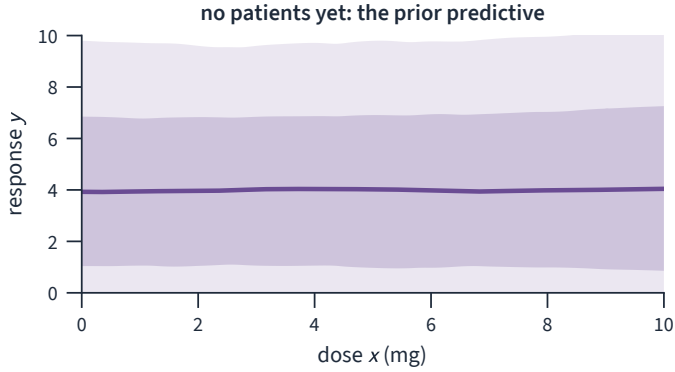




the weighted average of just these eight: an illustration

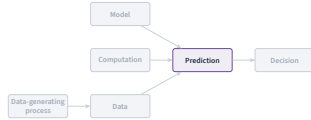
The prior predictive, setting by setting

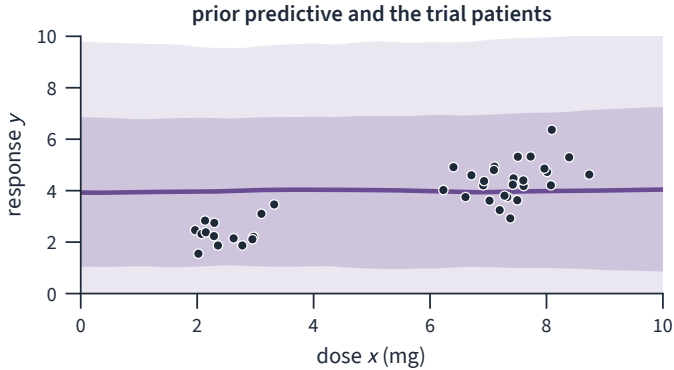




adding up all settings, with their weights: the prior predictive

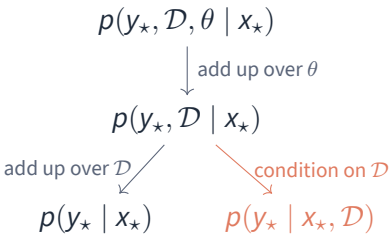
The data arrive

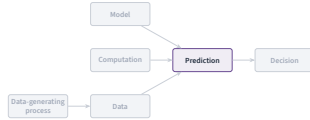




Before the data, almost any response is possible: we can do better

The posterior predictive



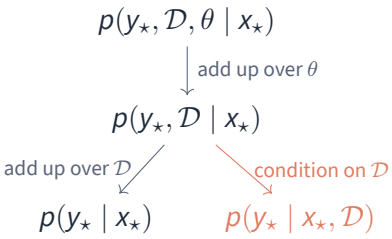


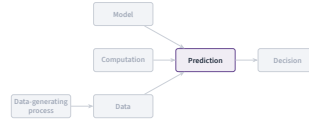
Now use the data: like Holmes, update the beliefs with what we observed

$$p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta) \underbrace{p(\theta \mid \mathcal{D})}_{\text{the posterior}} d\theta$$

$$\propto \int p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)\, d\theta$$

The posterior predictive





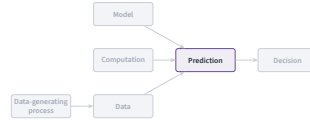
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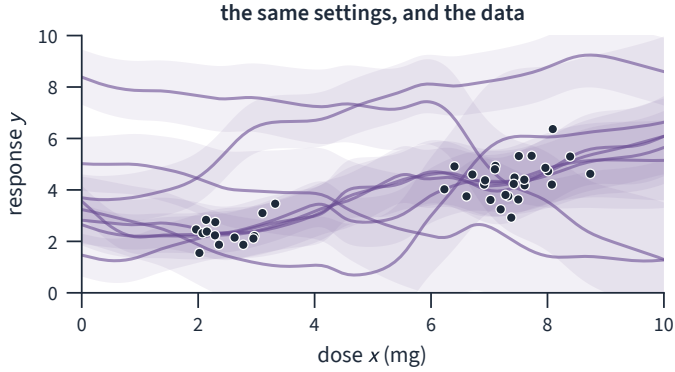
$$p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta) \underbrace{p(\theta \mid \mathcal{D})}_{\text{the posterior}} d\theta$$

$$\propto \int p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)\, d\theta$$

each setting's band, now weighted by $p(\theta)$ times how well it explains the data, $p(\mathcal{D} \mid \theta)/p(\mathcal{D})$

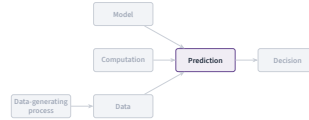
The posterior predictive, setting by setting

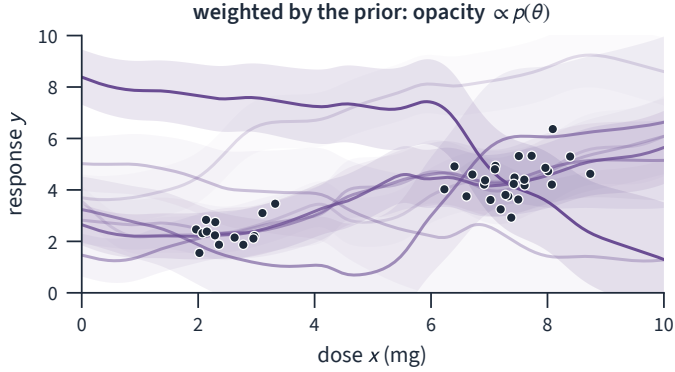




the same eight settings, with the trial patients

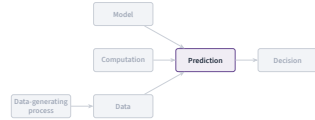
The posterior predictive, setting by setting

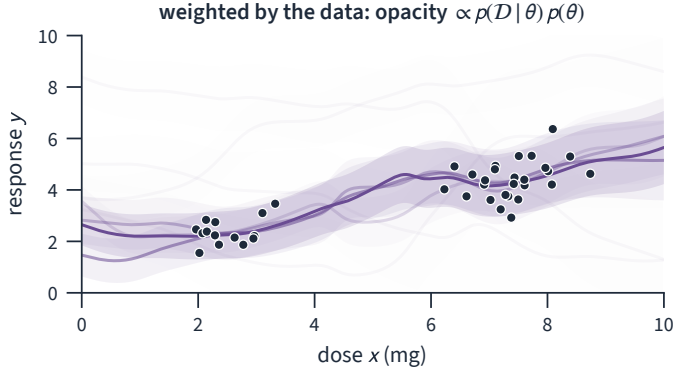




weighted by the prior

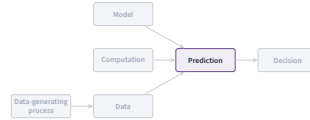
The posterior predictive, setting by setting

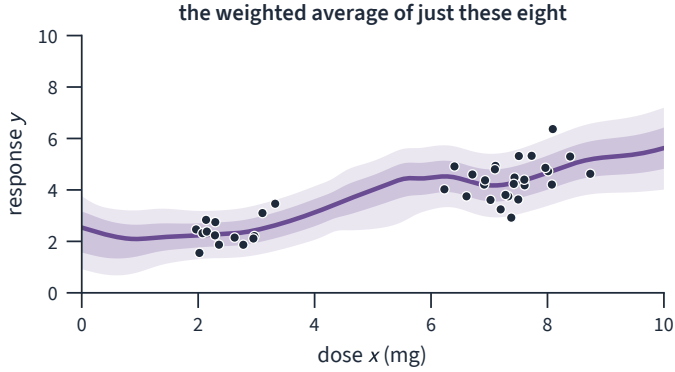




weighted by the prior and the data: settings that miss the data almost vanish

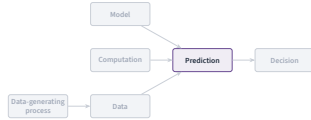
The posterior predictive, setting by setting

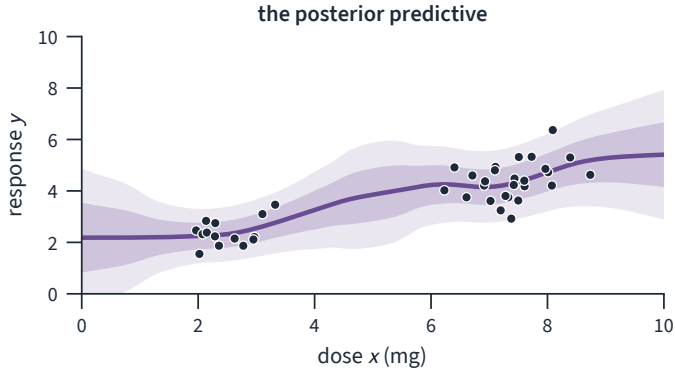




the weighted average of just these eight: an illustration

The posterior predictive, setting by setting

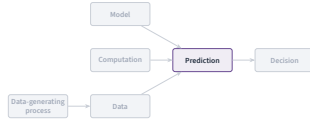




adding up all settings, with their new weights: the posterior predictive

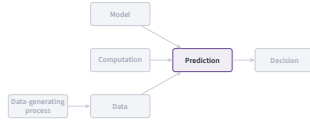
Prior and posterior predictive differ only in the weights

Before and after the data



$$p(y_\star \mid x_\star) = \int p(y_\star \mid x_\star, \theta)\, p(\theta)\, d\theta$$

Before and after the data

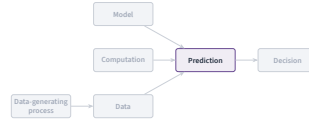


$$p(y_\star \mid x_\star) = \int p(y_\star \mid x_\star, \theta)\; p(\theta)\, d\theta$$

$$p(y_\star \mid x_\star, \mathcal{D}) \propto \int p(y_\star \mid x_\star, \theta)\; p(\mathcal{D} \mid \theta)\; p(\theta)\, d\theta$$

$\propto$: up to $p(\mathcal{D})$, which does not depend on $y_\star$

Before and after the data



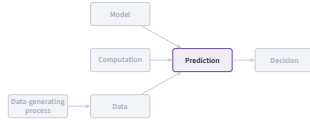
$$p(y_\star \mid x_\star) = \int p(y_\star \mid x_\star, \theta)\; p(\theta)\, d\theta$$

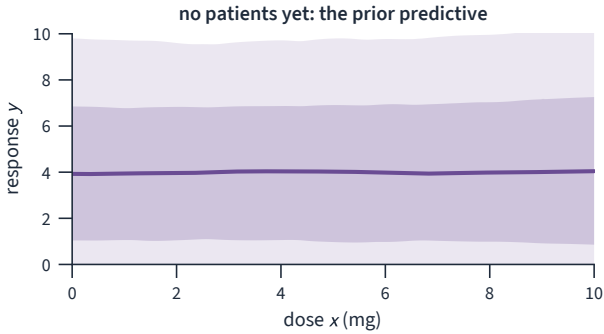
$$p(y_\star \mid x_\star, \mathcal{D}) \propto \int p(y_\star \mid x_\star, \theta)\; p(\mathcal{D} \mid \theta)\; p(\theta)\, d\theta$$

$\propto$: up to $p(\mathcal{D})$, which does not depend on $y_\star$

The same settings, now also weighted by how well they explain the data

More data, less uncertainty

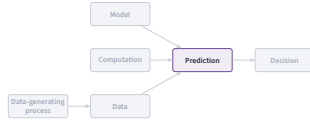


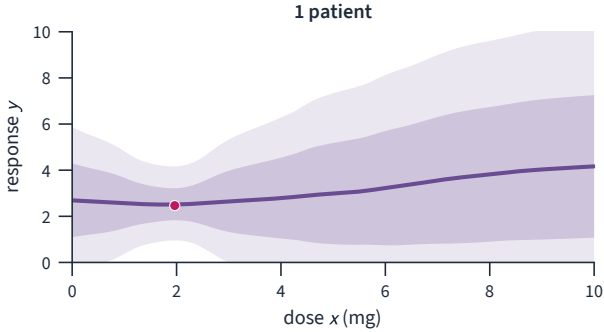


$$p(y_\star \mid x_\star) = \int p(y_\star \mid x_\star, \theta)\; p(\theta)\; d\theta$$

no patients yet: the prior predictive

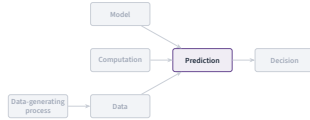
More data, less uncertainty

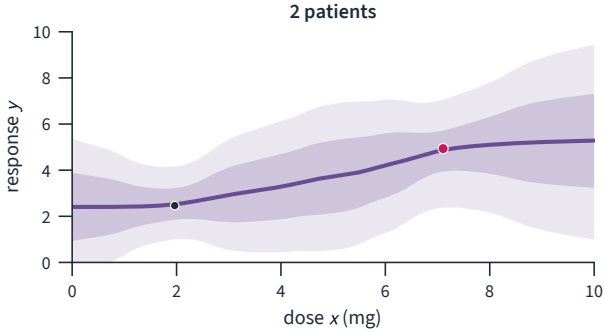




$$p(y_\star \mid x_\star, \mathcal{D}) \propto \int p(y_\star \mid x_\star, \theta) \prod_{i=1}^{1} p(y_i \mid x_i, \theta)\, p(\theta)\, d\theta$$

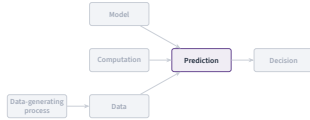
More data, less uncertainty

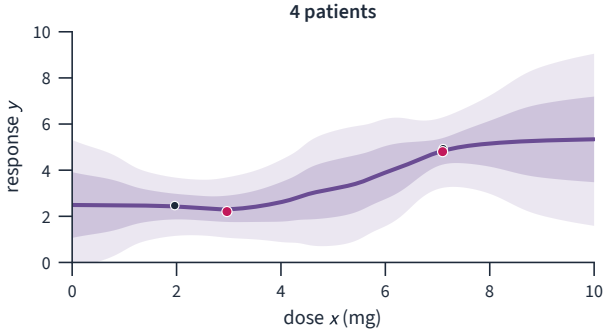




$$p(y_\star \mid x_\star, \mathcal{D}) \propto \int p(y_\star \mid x_\star, \theta) \prod_{i=1}^{2} p(y_i \mid x_i, \theta)\, p(\theta)\, d\theta$$

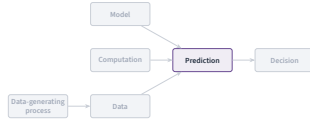
More data, less uncertainty

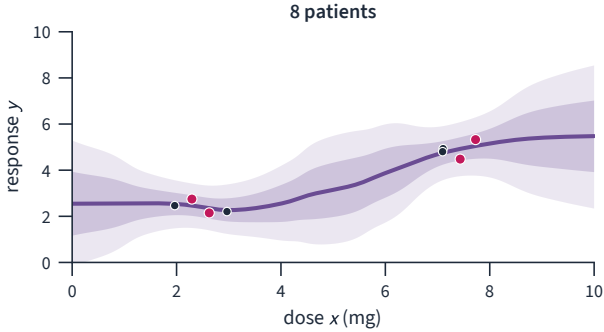




$$p(y_\star \mid x_\star, \mathcal{D}) \propto \int p(y_\star \mid x_\star, \theta) \prod_{i=1}^{4} p(y_i \mid x_i, \theta)\, p(\theta)\, d\theta$$

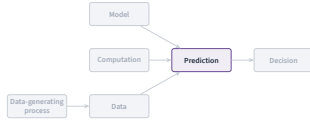
More data, less uncertainty

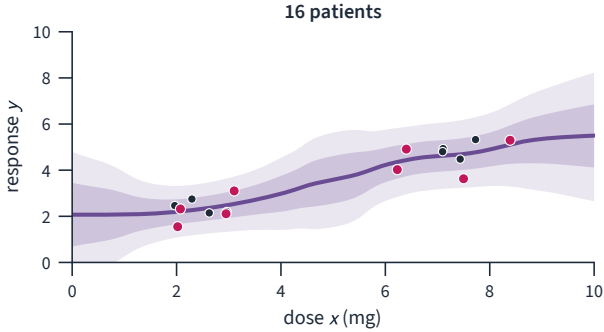




$$p(y_\star \mid x_\star, \mathcal{D}) \propto \int p(y_\star \mid x_\star, \theta) \prod_{i=1}^{8} p(y_i \mid x_i, \theta)\, p(\theta)\, d\theta$$

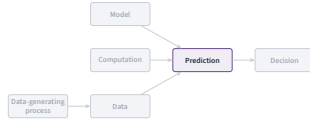
More data, less uncertainty

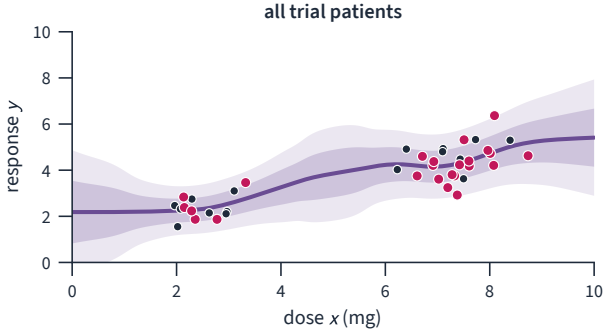




$$p(y_\star \mid x_\star, \mathcal{D}) \propto \int p(y_\star \mid x_\star, \theta) \prod_{i=1}^{16} p(y_i \mid x_i, \theta)\, p(\theta)\, d\theta$$

More data, less uncertainty

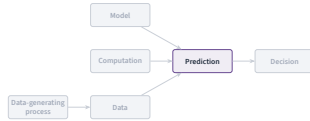


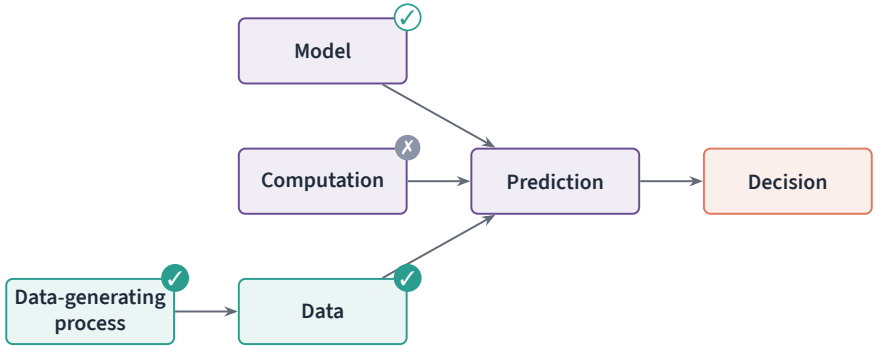


$$p(y_\star \mid x_\star, \mathcal{D}) \propto \int p(y_\star \mid x_\star, \theta) \prod_{i=1}^{40} p(y_i \mid x_i, \theta)\, p(\theta)\, d\theta$$

Each patient narrows the prediction nearby; far from all patients, it stays wide

What the prediction captures



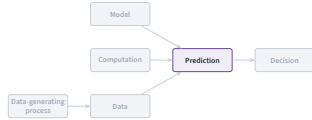


Captured: randomness in the process and limited data

Not captured: what we chose not to model, the model class, approximation error

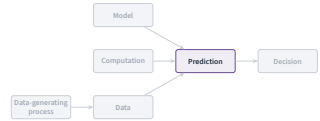
✓ captured ✓ partly ✗ not captured

Limitations



Every limitation comes from a modelling choice

Limitation: a wrong model



our choice

one model class *M*

consequence

a wrong model

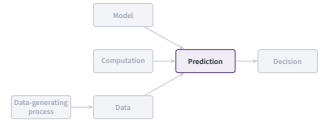
modelling it would let us

compare and check models

every prediction assumes the model is right: this network, a Gaussian spread, this prior

A posterior is only as good as its model

Limitation: unusual doses



our choice

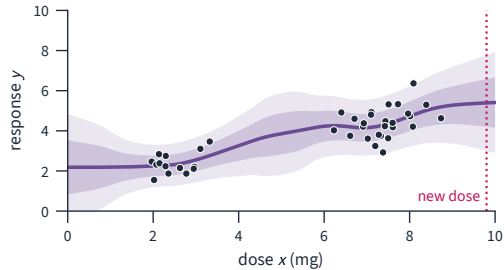
not modelling $p(x)$

consequence

unusual doses go unnoticed

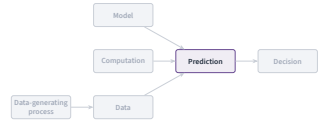
modelling it would let us

detect unusual doses

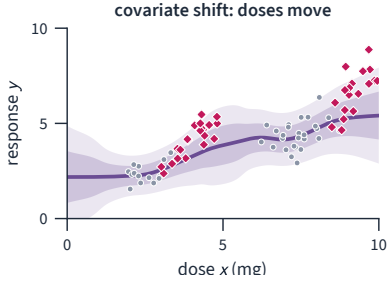


The model cannot tell us that a dose is unusual

Limitation: covariate shift



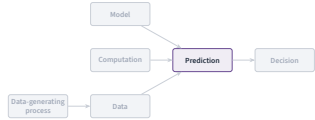
our choice	consequence	modelling it would let us
not modelling $p(x \mid t)$	covariate shift goes unnoticed	*detect* it



$p(y \mid x)$ **is unchanged, but new doses lie where data are scarce**

grey: trial patients pink: new patients at $t = 3$

Limitation: concept shift



our choice

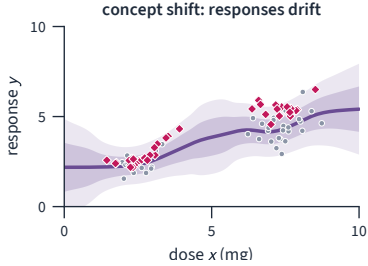
not modelling
$p(y \mid x, t)$

consequence

concept shift gives wrong predictions

modelling it would let us

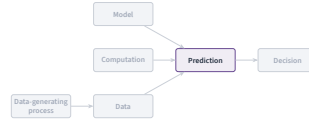
fix it, if t is recorded



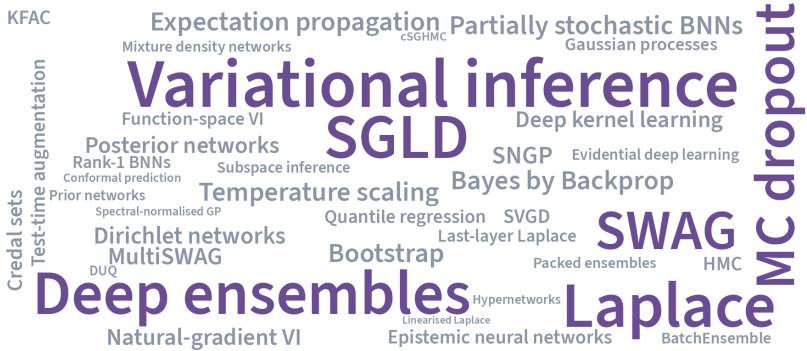
$p(y \mid x, t)$ has changed, so the predictions are wrong

grey: trial patients pink: new patients at $t = 3$

Computing the prediction

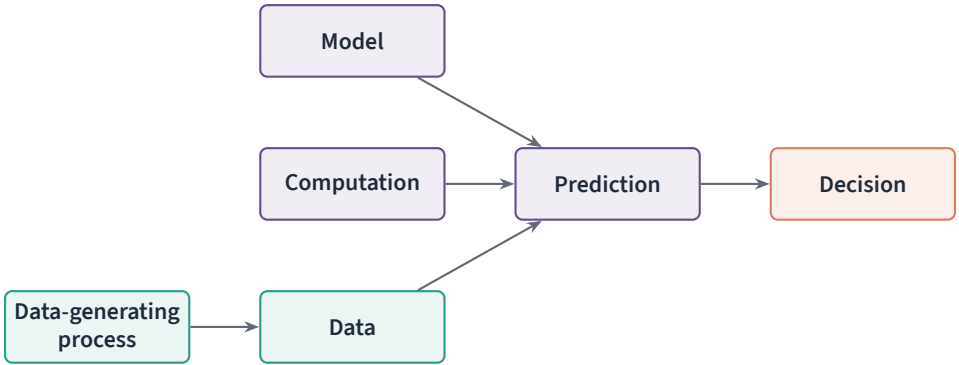


$$p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$$

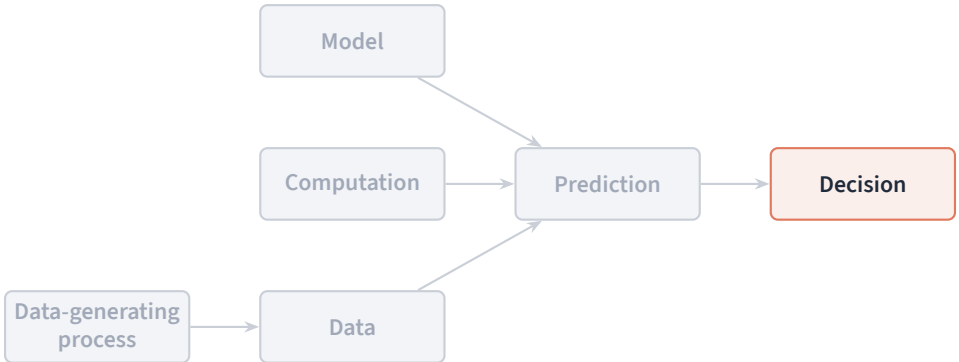


The methods from the first slide are different ways to compute this (Block 3)

Next: the decision

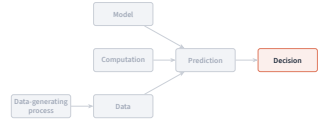


Next: the decision



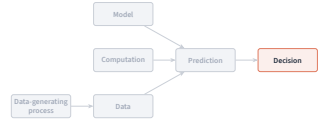
What do we truly care about, and what should we do?

How do we make decisions?



	rain	no rain
take an umbrella	1	1
leave it at home	10	0

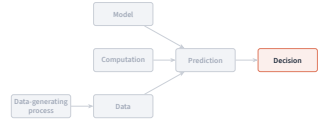
How do we make decisions?



	rain	no rain
take an umbrella	1	1
leave it at home	10	0

scenario $y_\star$ what happens; unknown when we decide

How do we make decisions?

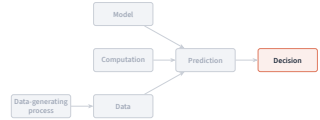


	rain	no rain
take an umbrella	1	1
leave it at home	10	0

scenario $y_\star$ what happens; unknown when we decide

action a what we choose

How do we make decisions?



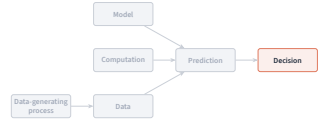
	rain	no rain
take an umbrella	1	1
leave it at home	10	0

scenario $y_\star$ what happens; unknown when we decide

action a what we choose

loss $\ell(a, y_\star)$ the cost of action a if the scenario is $y_\star$

How do we make decisions?



	rain	no rain
take an umbrella	1	1
leave it at home	10	0

scenario $y_\star$ what happens; unknown when we decide

action a what we choose

loss $\ell(a, y_\star)$ the cost of action a if the scenario is $y_\star$

If we knew it will rain, we would take the action with the smallest loss:

$$a_{\text{opt}} = \arg\min_a \ \ell(a, y_\star)$$ here: take the umbrella

Deciding with a prediction



What if we are uncertain about the scenario?

loss $\ell(a, y_\star)$

	rain	no rain
take	1	1
leave	10	0

$p(y_\star)$	
0.3	rain
0.7	no rain

Deciding with a prediction



What if we are uncertain about the scenario?

loss $\ell(a, y_\star)$

	rain	no rain
take	1	1
leave	10	0

$p(y_\star)$	
0.3	rain
0.7	no rain

	rain		**no rain**		**expected loss**
take	1	× 0.3 +	1	× 0.7 =	1.0
leave	10		0		3.0

$$a_{\text{opt}} = \arg\min_a \sum_{y_\star} \ell(a, y_\star)\, p(y_\star)$$

Deciding with a prediction



What if we are uncertain about the scenario?



$$a_{\text{opt}} = \arg\min_a \sum_{y_\star} \ell(a, y_\star)\, p(y_\star)$$

Take the umbrella; the loss we still expect, 1.0, is the uncertainty

Deciding under uncertainty



we don’t know the scenario $p(y_\star \mid x_\star, \mathcal{D})$

Deciding under uncertainty

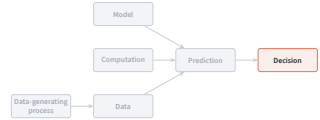


we don't know the scenario $p(y_\star \mid x_\star, \mathcal{D})$

average each action's loss $\int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$

We don't know the scenario, so we average over all likely scenarios

Deciding under uncertainty



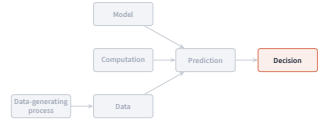
we don't know the scenario $p(y_\star \mid x_\star, \mathcal{D})$

average each action's loss $\int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$

choose the lowest $a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$

We don't know the scenario, so we average over all likely scenarios

Deciding under uncertainty



we don't know the scenario $p(y_\star \mid x_\star, \mathcal{D})$

average each action's loss $\int \ell(a, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$

choose the lowest $a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$

the loss we still expect **uncertainty** $= \int \ell(a_{\text{opt}}, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$

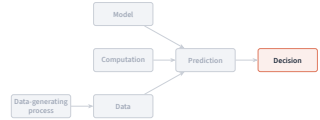
We don't know the scenario, so we average over all likely scenarios

Three kinds of decisions



we choose	example
a single value	report one number
a real-world action	approve a dose, or not
a whole distribution	report a prediction

Three kinds of decisions



we choose	example
a single value	report one number
a real-world action	approve a dose, or not
a whole distribution	report a prediction

Why report a single value?



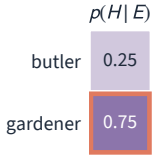
Holmes must accuse one suspect

	$p(H \mid E)$
butler	0.25
gardener	0.75

Why report a single value?



Holmes must accuse one suspect

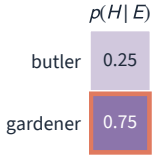


→ accuse the gardener

Why report a single value?

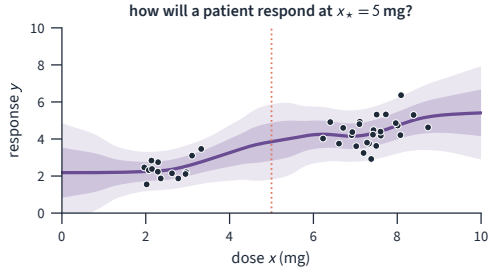


Holmes must accuse one suspect



→ accuse the gardener

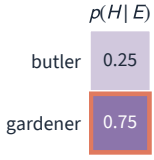
a doctor asks: one number, please



Why report a single value?

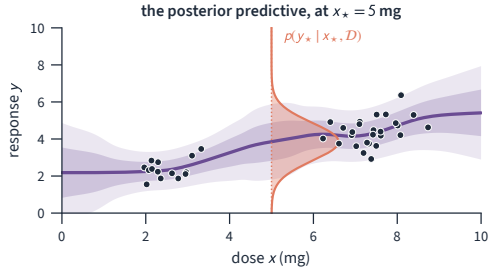


Holmes must accuse one suspect



→ accuse the gardener

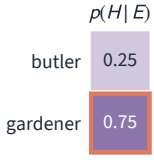
a doctor asks: one number, please



Why report a single value?

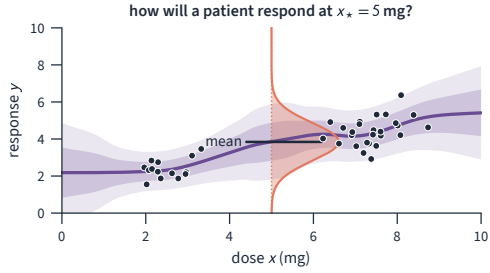


Holmes must accuse one suspect



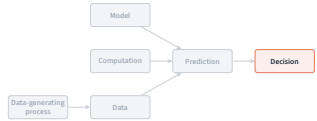
→ accuse the gardener

a doctor asks: one number, please

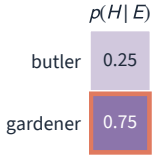


→ report, say, its mean

Why report a single value?

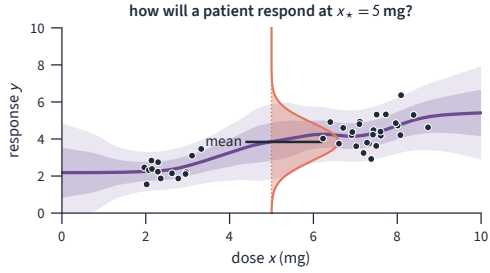


Holmes must accuse one suspect



→ accuse the gardener

a doctor asks: one number, please



→ report, say, its mean

Which single value? It depends on the loss

A single value: when we don't know the culprit



Holmes must accuse one suspect

loss $\ell(a, H)$
who did it?

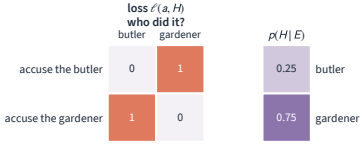
	butler	gardener
accuse the butler	0	1
accuse the gardener	1	0

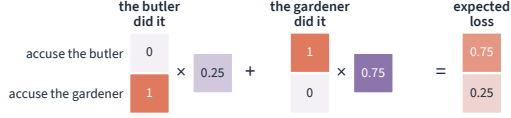
$p(H \mid E)$	
0.25	butler
0.75	gardener

A single value: when we don't know the culprit



Holmes must accuse one suspect



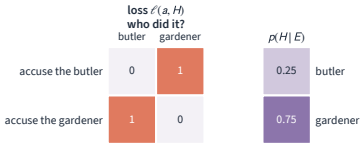


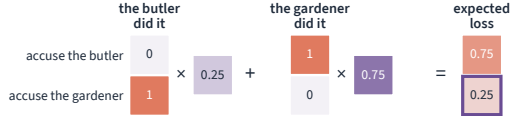
$$a_{\text{opt}} = \arg\min_a \sum_H \ell(a, H)\, p(H \mid E)$$

A single value: when we don't know the culprit



Holmes must accuse one suspect



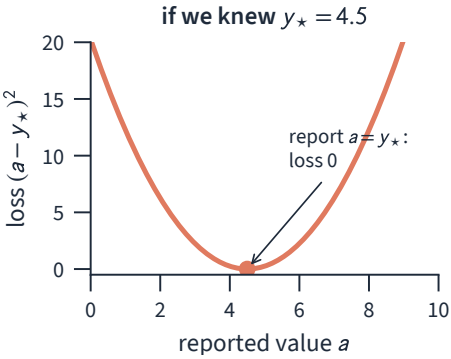


$$a_{\text{opt}} = \arg\min_a \sum_H \ell(a, H)\, p(H \mid E)$$

Accuse the gardener; the loss we still expect, 0.25, is the uncertainty

A single value: if we knew the outcome



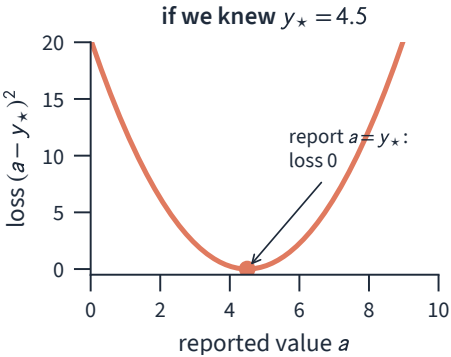


If we knew the outcome $y_\star$, we would choose the action with the smallest loss:

$$a_{\text{opt}} = \arg\min_a \ \ell(a, y_\star)$$

A single value: if we knew the outcome





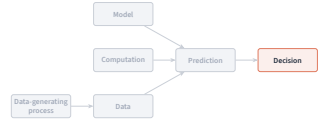
If we knew the outcome $y_\star$, we would choose the action with the smallest loss:

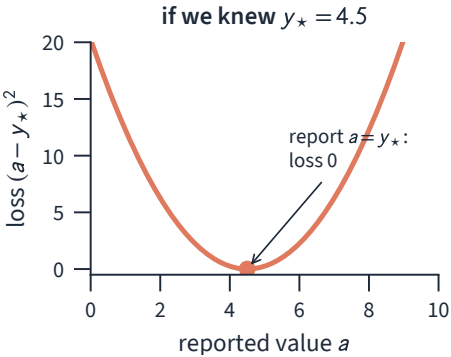
$$a_{\text{opt}} = \arg\min_a \ \ell(a, y_\star)$$

e.g. reporting a number, with squared loss:

$$a_{\text{opt}} = \arg\min_a \ (a - y_\star)^2 = y_\star$$

A single value: if we knew the outcome





If we knew the outcome $y_\star$, we would choose the action with the smallest loss:

$$a_{\text{opt}} = \arg\min_a \ \ell(a, y_\star)$$

e.g. reporting a number, with squared loss:

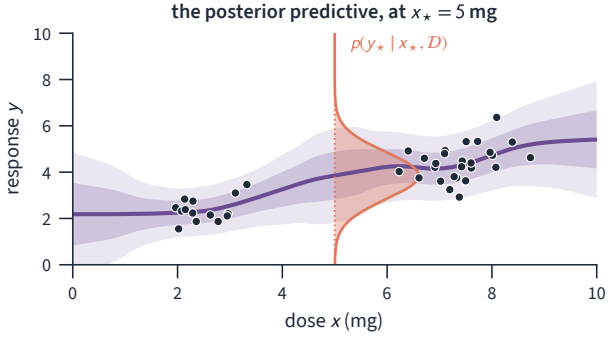
$$a_{\text{opt}} = \arg\min_a \ (a - y_\star)^2 = y_\star$$

Nothing is lost: no uncertainty

A single value: when we don't know the outcome

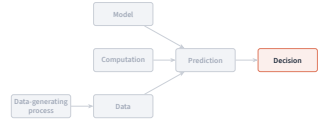


$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$$

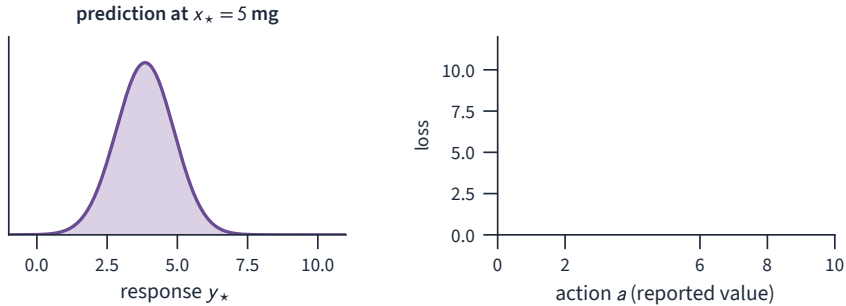


the posterior predictive at one dose $x_\star$

A single value: when we don't know the outcome

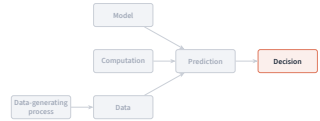


$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$$



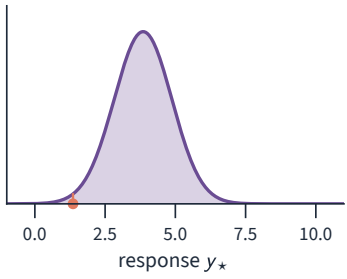
the prediction for $y_\star$ at that dose

A single value: when we don't know the outcome

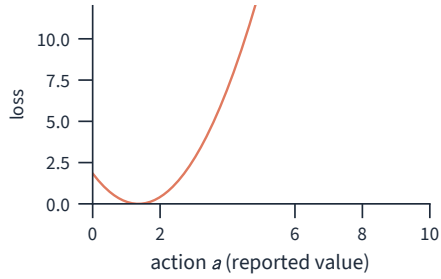


$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$$

prediction at $x_\star = 5$ mg

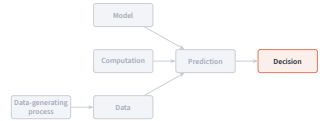


one loss curve per outcome: $(a - y_\star)^2$



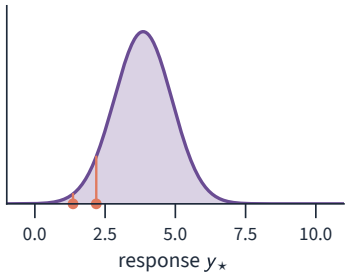
each possible outcome $y_\star$ gives its own loss curve over the actions

A single value: when we don't know the outcome

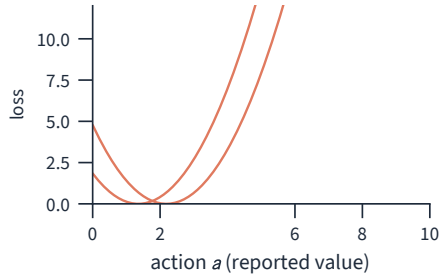


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prediction at $x_\star = 5$ mg

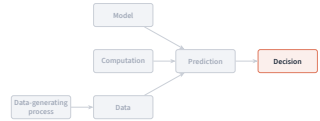


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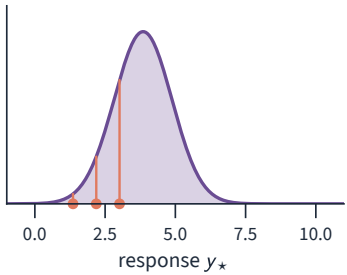
each possible outcome $y_\star$ gives its own loss curve over the actions

A single value: when we don't know the outcome

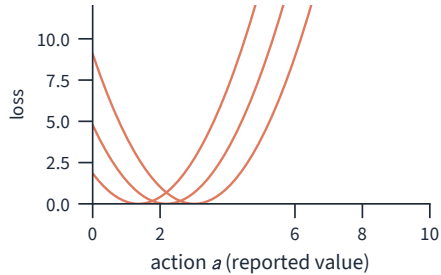


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prediction at $x_\star = 5$ mg

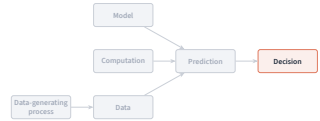


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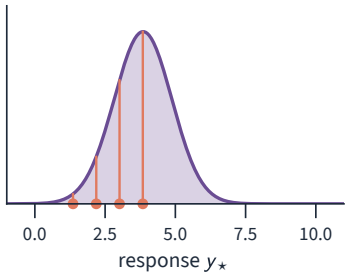
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A single value: when we don't know the outcome

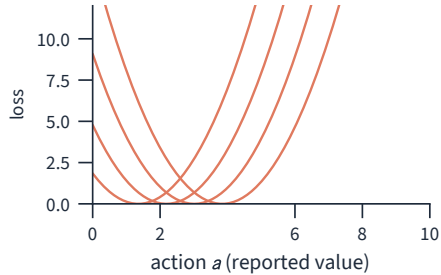


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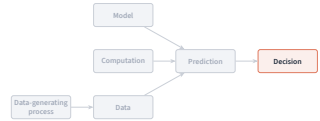


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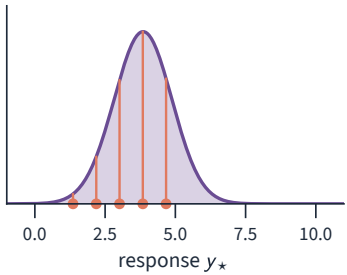
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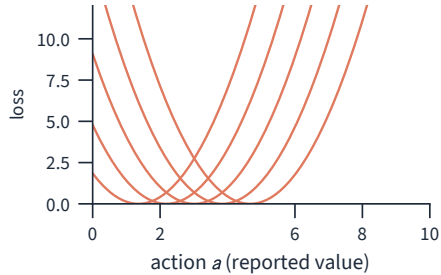


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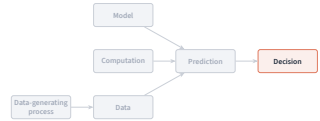


one loss curve per outcome: $(a - y_\star)^2$



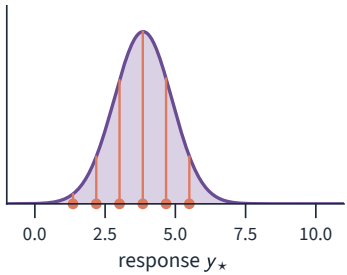
each possible outcome $y_\star$ gives its own loss curve over the actions

A single value: when we don't know the outcome

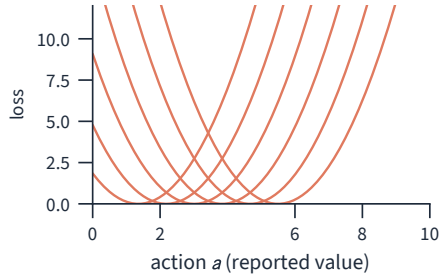


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prediction at $x_\star = 5$ mg

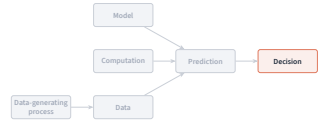


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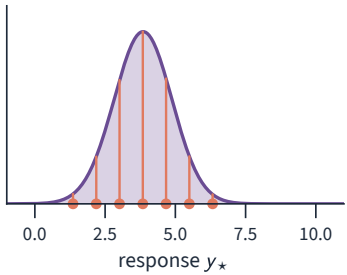
each possible outcome $y_\star$ gives its own loss curve over the actions

A single value: when we don't know the outcome

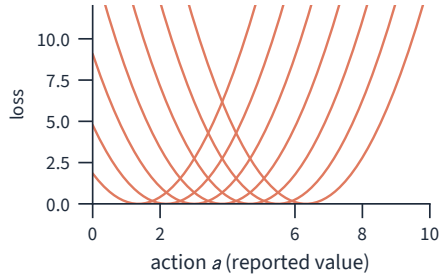


$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$$

prediction at $x_\star = 5$ mg

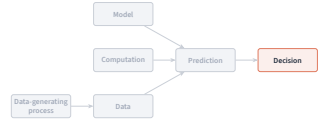


one loss curve per outcome: $(a - y_\star)^2$

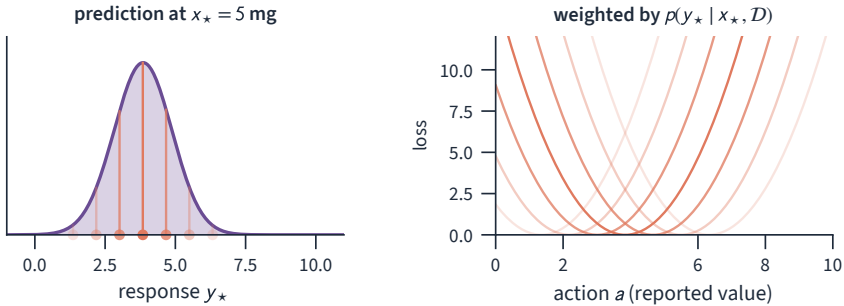


each possible outcome $y_\star$ gives its own loss curve over the actions

A single value: when we don't know the outcome

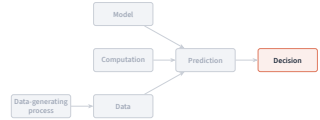


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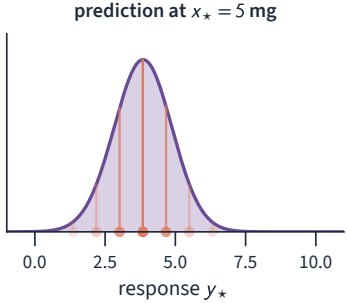


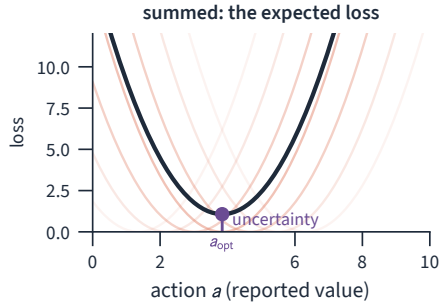
weight each curve by how likely its outcome is

A single value: when we don't know the outcome



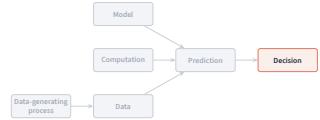
$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$$



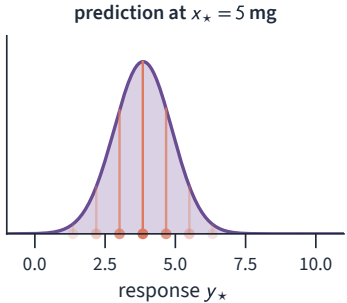


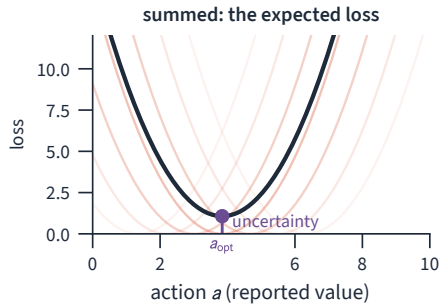
add them up: the expected loss; its lowest point is a_{opt}, its height the uncertainty

A single value: when we don't know the outcome



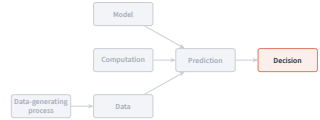
$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\; dy_\star$$





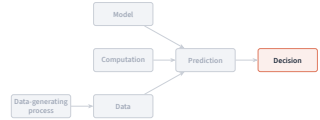
for squared loss: a_{opt} is the mean, and the uncertainty is the variance

Reporting a single value



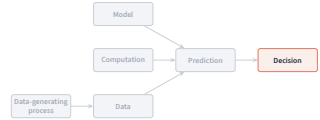
loss	optimal action	uncertainty
squared error	mean	variance

Reporting a single value



loss	optimal action	uncertainty
squared error	mean	variance
absolute error	median	mean abs. deviation

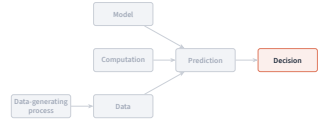
Reporting a single value



loss	optimal action	uncertainty
squared error	mean	variance
absolute error	median	mean abs. deviation
0–1	most likely value	$1 - \max$ probability

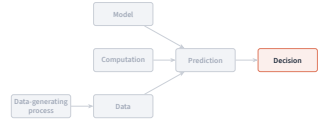
Each loss, its own optimal action and its own uncertainty

Three kinds of decisions



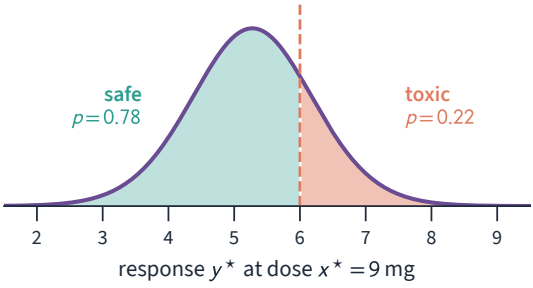
we choose	example
a single value	report one number
a real-world action	approve a dose, or not
a whole distribution	report a prediction

Three kinds of decisions



we choose	example
a single value	report one number
a real-world action	**approve a dose, or not**
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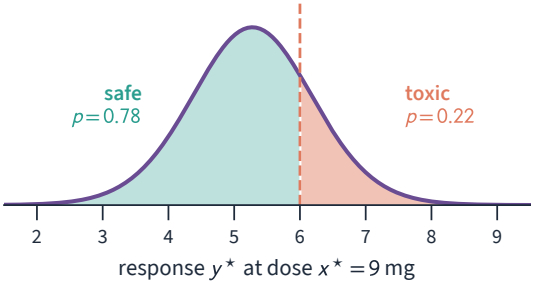
The scenario: the patient's response





The scenario: the patient's response





	toxic	safe
approve	**50**	0
reject	0	**1**
run more trials	0.75	0.75

Expected loss of each action



loss $\ell(a, y_\star)$

	toxic	safe
approve	50	0
reject	0	1
more trials	0.75	0.75

$p(y_\star \mid x_\star, \mathcal{D})$

p	toxic
$1 - p$	safe

Expected loss of each action



loss $\ell(a, y_\star)$

	toxic	safe
approve	50	0
reject	0	1
more trials	0.75	0.75

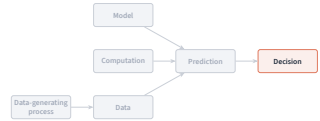
$p(y_\star \mid x_\star, \mathcal{D})$

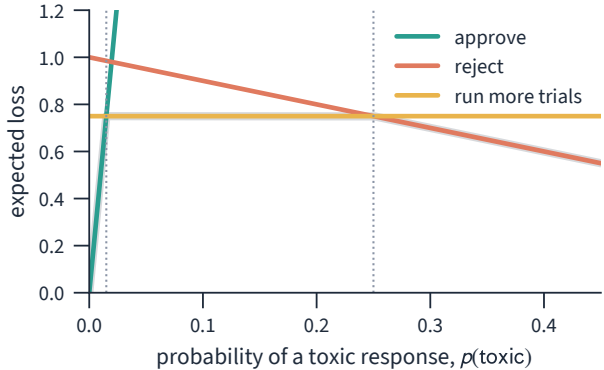
p	toxic
$1-p$	safe

	toxic			safe			expected loss
approve	50			0			$50p$
reject	0	$\times$	p	$+$ 1	$\times$	$1-p$	$=$ $1-p$
more trials	0.75			0.75			0.75

different $p = p(\text{toxic})$, different expected losses

Expected loss of each action



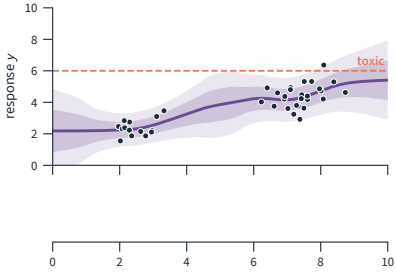


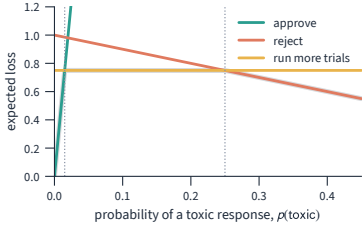
Approve if $p(\text{toxic}) < 1.5\%$, reject if $p(\text{toxic}) > 25\%$, otherwise run more trials

Should we approve this dose?



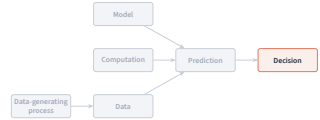
PREDICT Which action wins in the gap?

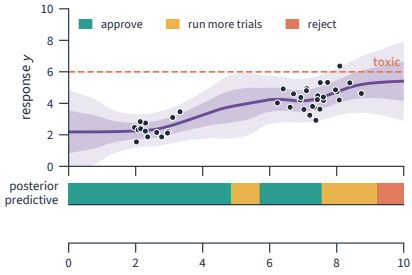


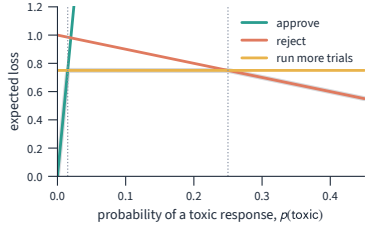


Whether more trials pay off depends on how much of the expected loss more data could remove: F. Bickford Smith et al. (2024). Rethinking aleatoric and epistemic uncertainty. arXiv:2412.20892.

Should we approve this dose?

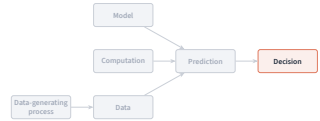


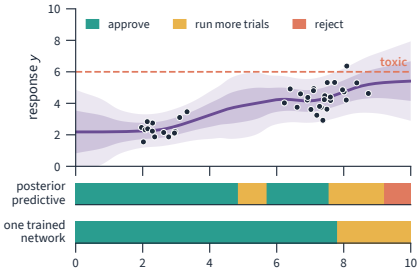


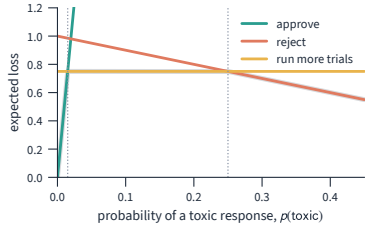


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Should we approve this dose?





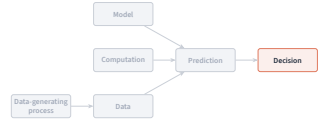


Uncertainty changes the decision

one trained network: approve posterior predictive: run more trials

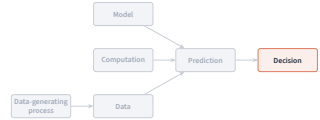
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Three kinds of decisions



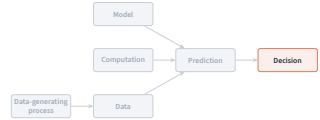
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Three kinds of decisions

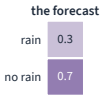


we choose	example
a single value	report one number
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a whole distribution	**report a prediction**

Why report a whole distribution?



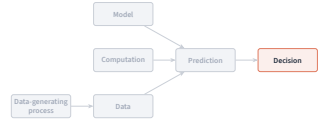
You provide a prediction $p(y_\star \mid x_\star, \mathcal{D})$: say, a 30% chance of rain.



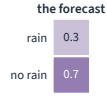
Different people have different losses:



Why report a whole distribution?



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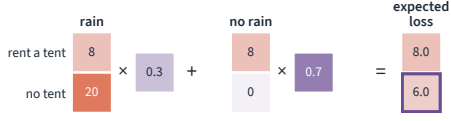


Different people have different losses:

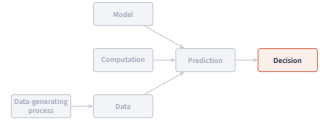
you



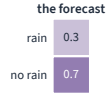
an event organiser



Why report a whole distribution?



You provide a prediction $p(y_\star \mid x_\star, \mathcal{D})$: say, a 30% chance of rain.

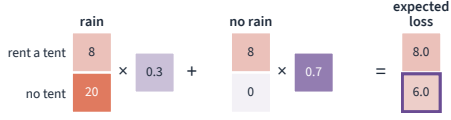


Different people have different losses:

you



an event organiser



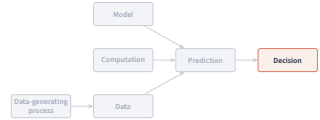
When others decide, report the distribution, and let them apply their own losses

Reporting a distribution is a decision, but for which loss?



action	report a distribution $q(y_\star)$
scenario	the outcome $y_\star$
belief	$p(y_\star \mid x_\star, \mathcal{D})$

Reporting a distribution is a decision, but for which loss?



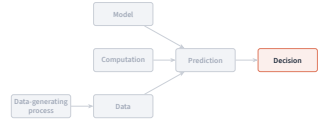
action	report a distribution $q(y_\star)$
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Which losses make our belief the best report?

$$\arg\min_q \int \ell(q, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\, dy_\star \;=\; p(y_\star \mid x_\star, \mathcal{D})$$

such losses are called *proper*

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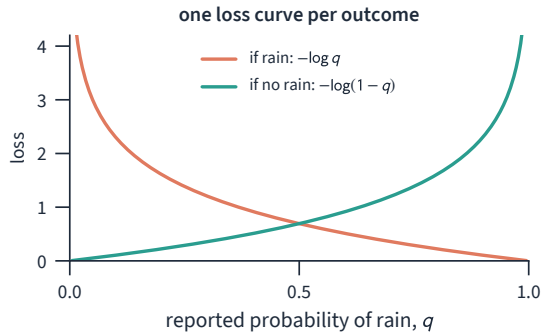
The log loss is one: $\ell(q, y_\star) = -\log q(y_\star)$

Reporting a whole distribution: the forecaster



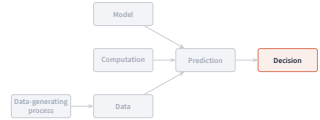
Which $q(\text{rain})$ should the forecaster report? The loss: the log loss

	rain	no rain
report $q(y_\star)$	q	$1-q$
belief $p(y_\star)$	0.3	0.7
loss $\ell(q, y_\star)$	$-\log q$	$-\log(1-q)$



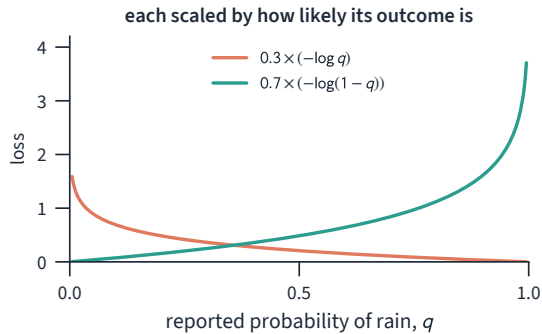
one loss curve per outcome

Reporting a whole distribution: the forecaster



Which $q(\text{rain})$ should the forecaster report? The loss: the log loss

	rain	no rain
report $q(y_\star)$	q	$1-q$
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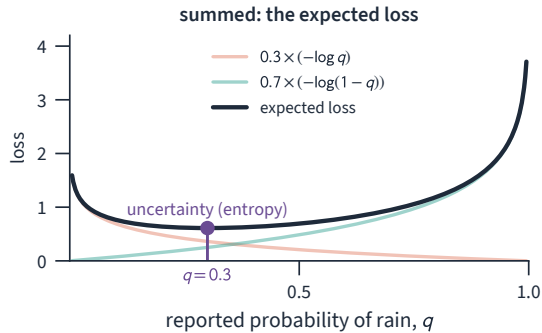
each scaled by how likely its outcome is

Reporting a whole distribution: the forecaster



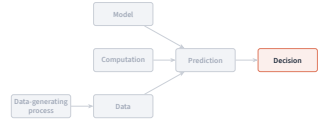
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loss $\ell(q, y_\star)$	$-\log q$	$-\log(1-q)$



Lowest at $q = 0.3$, our belief; the loss we still expect is the uncertainty

Discussion: what is uncertainty?



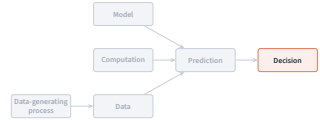
$$\text{uncertainty} = \int \ell(a_{\text{opt}}, y_\star)\; p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$$

it depends on the loss *and* on the belief



- for a fixed loss, does it make sense that the uncertainty changes with the belief?
- for a fixed belief, does it make sense that the uncertainty changes with the loss?

Evaluation: our beliefs and the true process



our beliefs

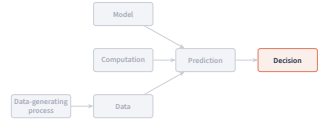
optimal action

$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$$

uncertainty

$$\int \ell(a_{\text{opt}}, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$$

Evaluation: our beliefs and the true process



our beliefs

optimal action

$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$$

uncertainty

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the true process

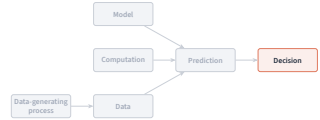
optimal action

$$a_{\text{opt}}^{\text{true}} = \arg\min_a \int \ell(a, y_\star)\, p_{\text{true}}(y_\star \mid x_\star)\, dy_\star$$

irreducible uncertainty

$$\int \ell(a_{\text{opt}}^{\text{true}}, y_\star)\, p_{\text{true}}(y_\star \mid x_\star)\, dy_\star$$

Evaluation: our beliefs and the true process



our beliefs

optimal action

$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$$

uncertainty

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the true process

optimal action

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irreducible uncertainty

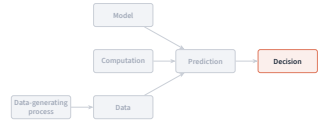
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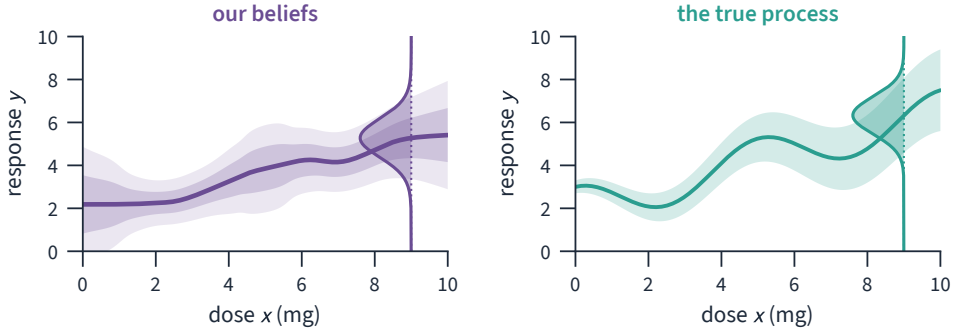
evaluation

$$\int \ell(a_{\text{opt}}, y_\star)\, p_{\text{true}}(y_\star \mid x_\star)\, dy_\star$$

our action, the true process's loss; in practice, averaged over held-out patients

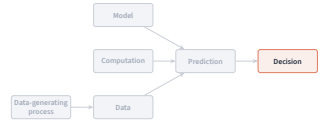
Evaluation, step by step

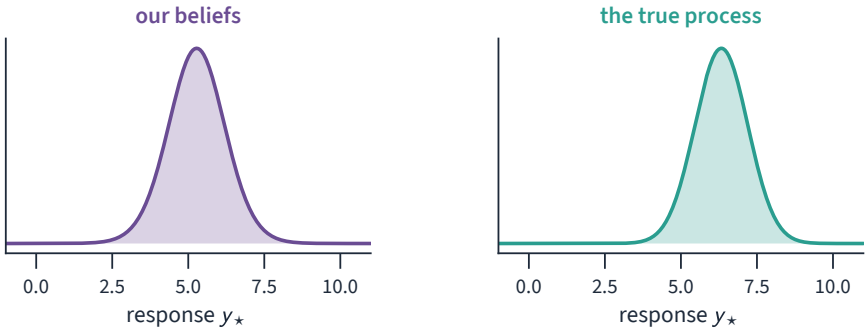




our predictive and the true process, at one dose $x_\star$

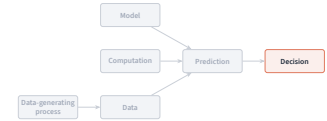
Evaluation, step by step



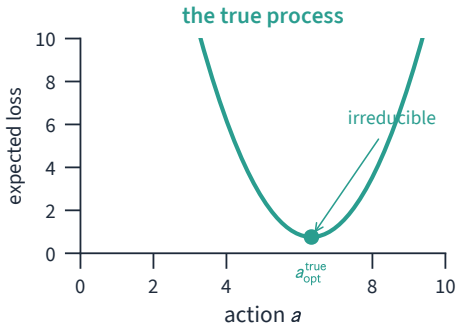


the two distributions over the response

Evaluation, step by step

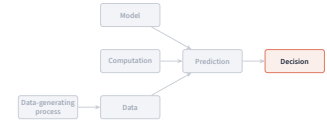


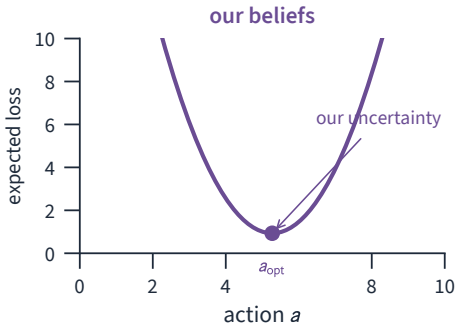


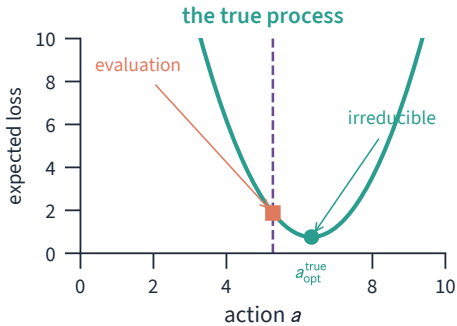


each gives an expected loss curve and an optimal action

Evaluation, step by step

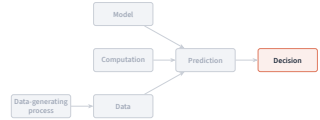






evaluation: our optimal action, scored with nature's expected loss

Evaluation: the whole picture



our beliefs

optimal action

$$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$$

uncertainty

$$\int \ell(a_{\text{opt}}, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$$

the true process

optimal action

$$a^{\text{true}}_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\, p_{\text{true}}(y_\star \mid x_\star)\, dy_\star$$

irreducible uncertainty

$$\int \ell(a^{\text{true}}_{\text{opt}}, y_\star)\, p_{\text{true}}(y_\star \mid x_\star)\, dy_\star$$

evaluation

$$\int \ell(a_{\text{opt}}, y_\star)\, p_{\text{true}}(y_\star \mid x_\star)\, dy_\star$$

our action, the true process's loss; in practice, averaged over held-out patients

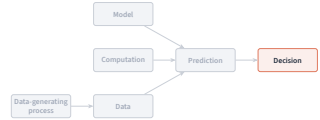
What evaluation tells us



our uncertainty is the model's forecast of the evaluation

if the two disagree on held-out patients, the model is over- or underconfident

What evaluation tells us

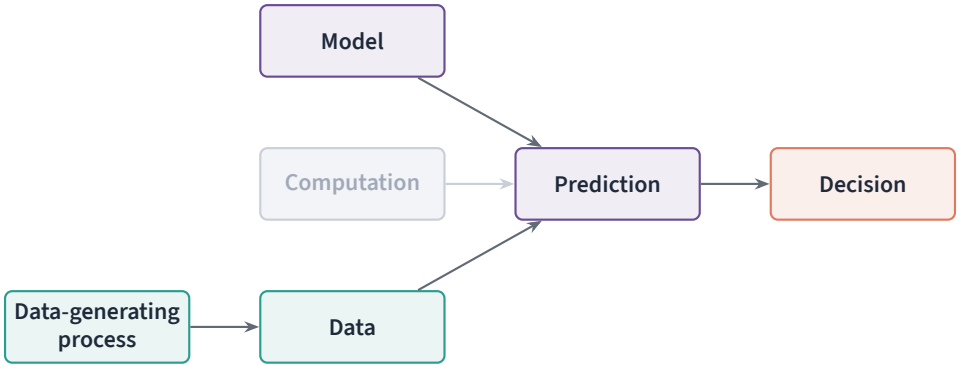


our uncertainty is the model's forecast of the evaluation
if the two disagree on held-out patients, the model is over- or underconfident

the evaluation can never beat the irreducible loss
the gap comes from limited data, our model choices, and computation; the irreducible part comes from the true process

Evaluate with the loss you care about, on data the model has not seen

Where we are


Model
Computation
Prediction
Decision
Data-generating process
Data

Break